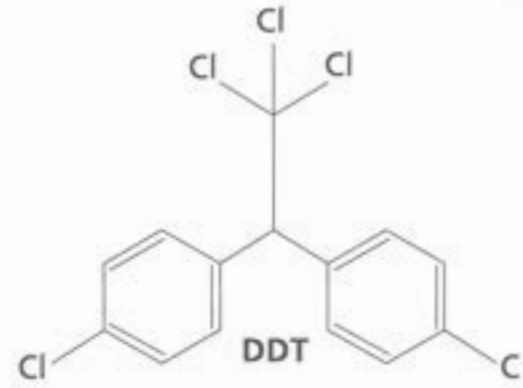
DDT: the Peregrine Falcon case

A chlorinated hydrocarbon compound known as *dichloro diphenyl trichloroethane*, or DDT, provided one of the earliest examples of the effects of biomagnification. DDT is a colourless, odourless and very effective insecticide that first came into use in the 1940s.



DDT degrades slowly, with a half-life of 15 years, so it is considered a non-degradable substance. It is not water-soluble but, being fat-soluble, is taken up by fatty tissue within the body. DDT illustrated the sub-lethal effects of biomagnification. *Sub-lethal effects* are those that *indirectly* cause death.

1 Define *insecticide*.

2 What is meant by *half-life*?

The peregrine falcon is one of a number of predatory birds belonging to the raptor family. During the decade from the mid-1950s to mid-1960s, the numbers of surviving falcon chicks dropped dramatically. Tests revealed that the parent falcons contained high levels of DDT yet they appeared able to cope with it – so what was the problem? Upon inspecting the nests, it was observed that the shells of the falcon eggs were unnaturally thin. During normal incubation, the parents could not avoid breaking the paper-thin shells. It became obvious that DDT levels were directly affecting the hen's ability to produce a thick enough shell.

DDT did not directly kill the chicks, but indirectly killed them by preventing a strong enough eggshell in which to develop. Similar effects have been observed on Bald Eagles and other predatory birds. The prey does not contain sufficient levels of DDT to harm them but the birds accumulate the toxin with each new feed. Fortunately, the establishment of captive breeding programs has enabled the peregrine falcon to make a comeback and the use of DDT has since been banned in most developed countries.

3 Suggest how the peregrine falcons accumulated so much DDT in their tissues.

DDT is a 'broad spectrum' pesticide, meaning that it affects a broad range of pest species and not just the target species. There have been many documented cases where the pesticide was sprayed on crops to destroy insects that ate the crop, for example locusts. Birds and lizards swarmed on the easy meals that lay dying on the ground. Soon after the birds and lizards began to die from DDT poisoning, as well as those that preyed on them, such as cats or larger birds and reptiles – another example of biomagnification at work.

At the same time other insect species moved in to eat the crops. Normally these species would be kept under control by wasps, birds and other predators but, since DDT also killed those *non-target species*, the new arrivals were free to replace the original pests. Compound this with a few of the surviving original pests developing resistance, and the whole cycle repeats itself, minus some helpful, naturally occurring predators.

4 What is meant by a *'broad spectrum'* pesticide?

5 Explain why the fact that DDT is fat-soluble makes it so dangerous to vertebrates.

Integrated pest management: DDT and pest control

To *integrate* means to incorporate a number of factors, or to form a whole unit out of different component parts. *Integrated Pest Management (IPM)* incorporates various techniques to produce a management strategy that is the safest, most efficient possible for a given situation. After the effects of DDT became apparent, IPM strategies were recognised as the sensible alternative.

Research

6 Find out techniques that are used in IPM.

7 Read 'Silent Spring' (Rachel Carson, 1962) and comment on its relevance to this topic.

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